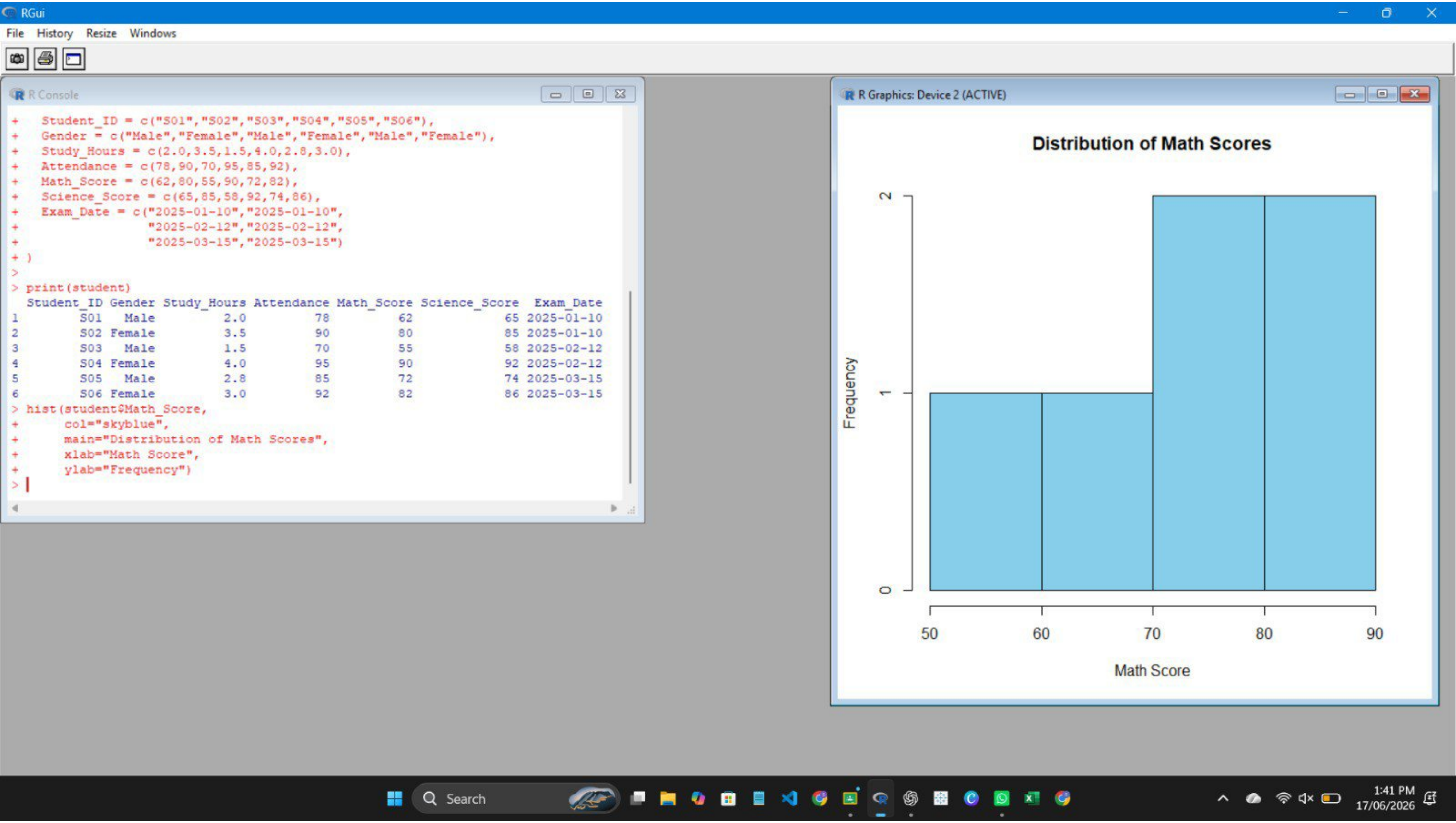
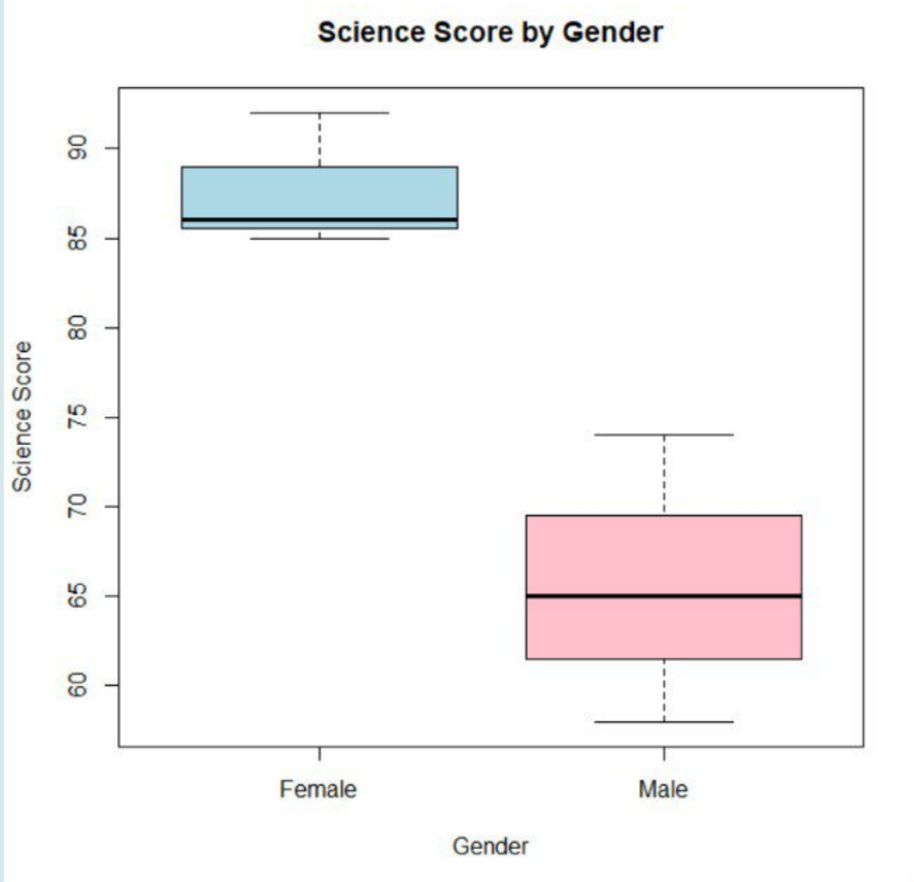

RGui
File History Resize Windows
R Console
+ Student_ID = c("S01","S02","S03","S04","S05","S06"),
+ Gender = c("Male","Female","Male","Female","Male","Female"),
+ Study_Hours = c(2.0,3.5,1.5,4.0,2.8,3.0),
+ Attendance = c(78,90,70,95,85,92),
+ Math_Score = c(62,80,55,90,72,82),
+ Science_Score = c(65,85,58,92,74,86),
+ Exam_Date = c("2025-01-10","2025-01-10",
+ "2025-02-12","2025-02-12",
+ "2025-03-15","2025-03-15")
+)
>
> print(student)
Student_ID Gender Study_Hours Attendance Math_Score Science_Score Exam_Date
1 S01 Male 2.0 78 62 65 2025-01-10
2 S02 Female 3.5 90 80 85 2025-01-10
3 S03 Male 1.5 70 55 58 2025-02-12
4 S04 Female 4.0 95 90 92 2025-02-12
5 S05 Male 2.8 85 72 74 2025-03-15
6 S06 Female 3.0 92 82 86 2025-03-15
> hist(student$Math_Score,
+ col="skyblue",
+ main="Distribution of Math Scores",
+ xlab="Math Score",
+ ylab="Frequency")
>
R Graphics: Device 2 (ACTIVE)
Distribution of Math Scores
Frequency
0
1
2
50
60
70
80
90
Math Score
Search
1:41 PM
17/06/2026

File Edit View Misc Packages Windows Help

R Console

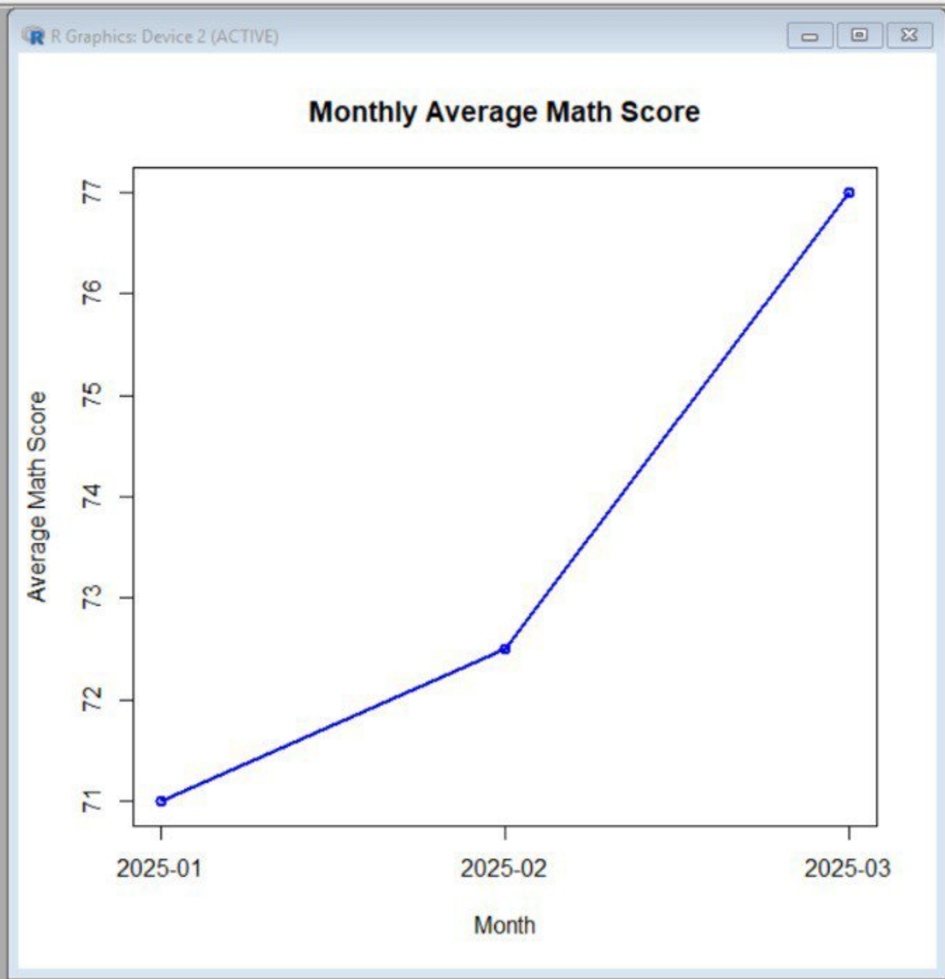
```
+   Exam_Date = c("2025-01-10","2025-01-10",
+                 "2025-02-12","2025-02-12",
+                 "2025-03-15","2025-03-15")
+ )
>
> print(student)
  Student_ID Gender Study_Hours Attendance Math_Score Science_Score  Exam_Date
1        S01   Male         2.0         78         62            65 2025-01-10
2        S02 Female         3.5         90         80            85 2025-01-10
3        S03   Male         1.5         70         55            58 2025-02-12
4        S04 Female         4.0         95         90            92 2025-02-12
5        S05   Male         2.8         85         72            74 2025-03-15
6        S06 Female         3.0         92         82            86 2025-03-15
> hist(student$Math_Score,
+      col="skyblue",
+      main="Distribution of Math Scores",
+      xlab="Math Score",
+      ylab="Frequency")
> boxplot(Science_Score ~ Gender,
+         data=student,
+         col=c("lightblue","pink"),
+         main="Science Score by Gender",
+         xlab="Gender",
+         ylab="Science Score")
>
```

R Graphics: Device 2 (ACTIVE)



R Console

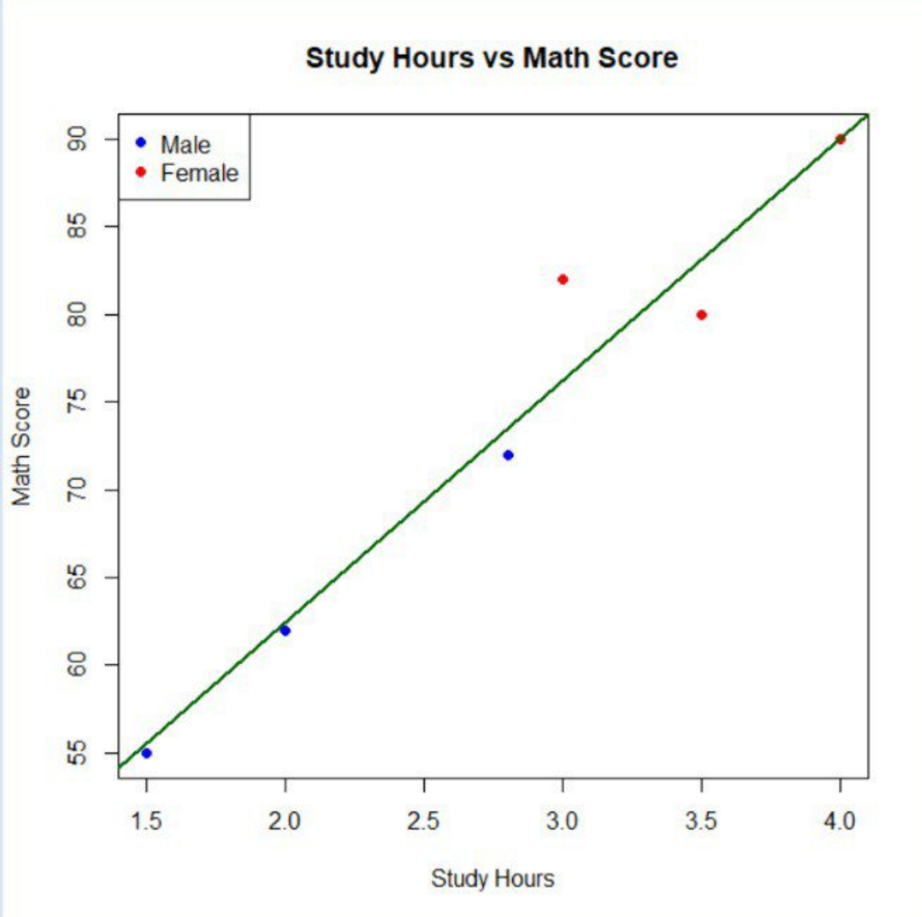
```
>
> monthly_avg <- aggregate(
+   Math_Score ~ Month,
+   data=student,
+   mean
+ )
>
> print(monthly_avg)
    Month Math_Score
1 2025-01       71.0
2 2025-02       72.5
3 2025-03       77.0
> plot(monthly_avg$Math_Score,
+      type="o",
+      col="blue",
+      lwd=2,
+      xaxt="n",
+      xlab="Month",
+      ylab="Average Math Score",
+      main="Monthly Average Math Score")
>
> axis(1,
+      at=1:nrow(monthly_avg),
+      labels=monthly_avg$Month)
>
```

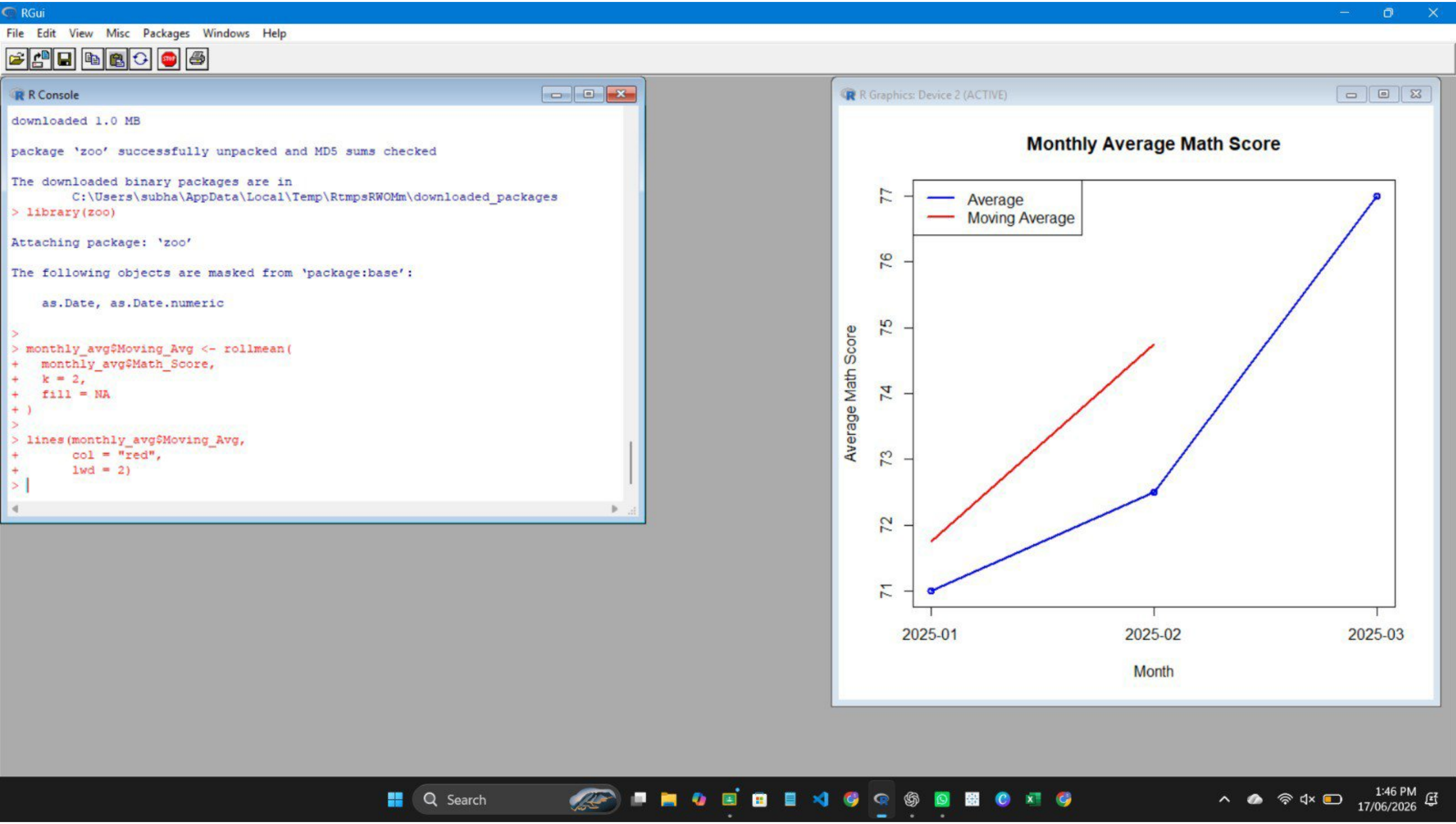


R Console

```
+       col="red",
+       lwd=2)
>
> legend("topleft",
+        legend=c("Average","Moving Average"),
+        col=c("blue","red"),
+        lwd=2)
> plot(student$Study_Hours,
+      student$Math_Score,
+      col=ifelse(student$Gender=="Male","blue","red"),
+      pch=19,
+      xlab="Study Hours",
+      ylab="Math Score",
+      main="Study Hours vs Math Score")
>
> abline(lm(Math_Score~Study_Hours,
+           data=student),
+        col="darkgreen",
+        lwd=2)
>
> legend("topleft",
+        legend=c("Male","Female"),
+        col=c("blue","red"),
+        pch=19)
>
```

R Graphics: Device 2 (ACTIVE)




RGui
File Edit View Misc Packages Windows Help
R Console
downloaded 1.0 MB
package ‘zoo’ successfully unpacked and MD5 sums checked
The downloaded binary packages are in
C:\Users\subha\AppData\Local\Temp\RtmpsRWOMm\downloaded_packages
> library(zoo)
Attaching package: ‘zoo’
The following objects are masked from ‘package:base’:
as.Date, as.Date.numeric
>
> monthly_avg$Moving_Avg <- rollmean(
+ monthly_avg$Math_Score,
+ k = 2,
+ fill = NA
+)
>
> lines(monthly_avg$Moving_Avg,
+ col = "red",
+ lwd = 2)
>
R Graphics: Device 2 (ACTIVE)
Monthly Average Math Score
Average
Moving Average
Average Math Score
71
72
73
74
75
76
77
2025-01
2025-02
2025-03
Month
Search
1:46 PM
17/06/2026

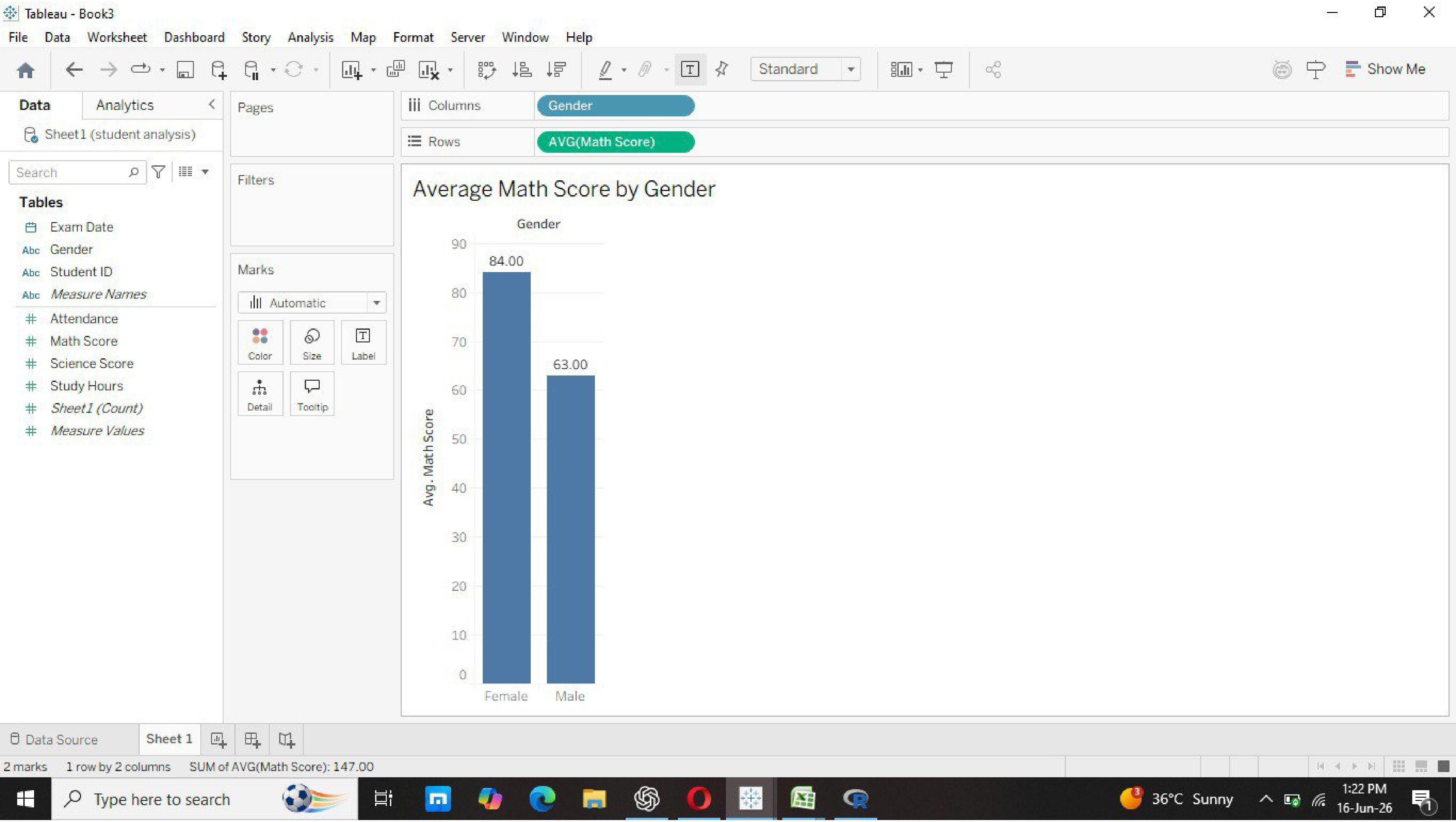


Tableau - Book3
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Show Me
Data
Analytics
Sheet1 (student analysis)
Search
Tables
Exam Date
Gender
Student ID
Measure Names
Attendance
Math Score
Science Score
Study Hours
Sheet1 (Count)
Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Columns
Gender
Rows
AVG(Math Score)
Average Math Score by Gender
Gender
84.00
63.00
Avg. Math Score
Female
Male
Data Source
Sheet 1
2 marks
1 row by 2 columns
SUM of AVG(Math Score): 147.00
Type here to search
36°C Sunny
1:22 PM
16-Jun-26

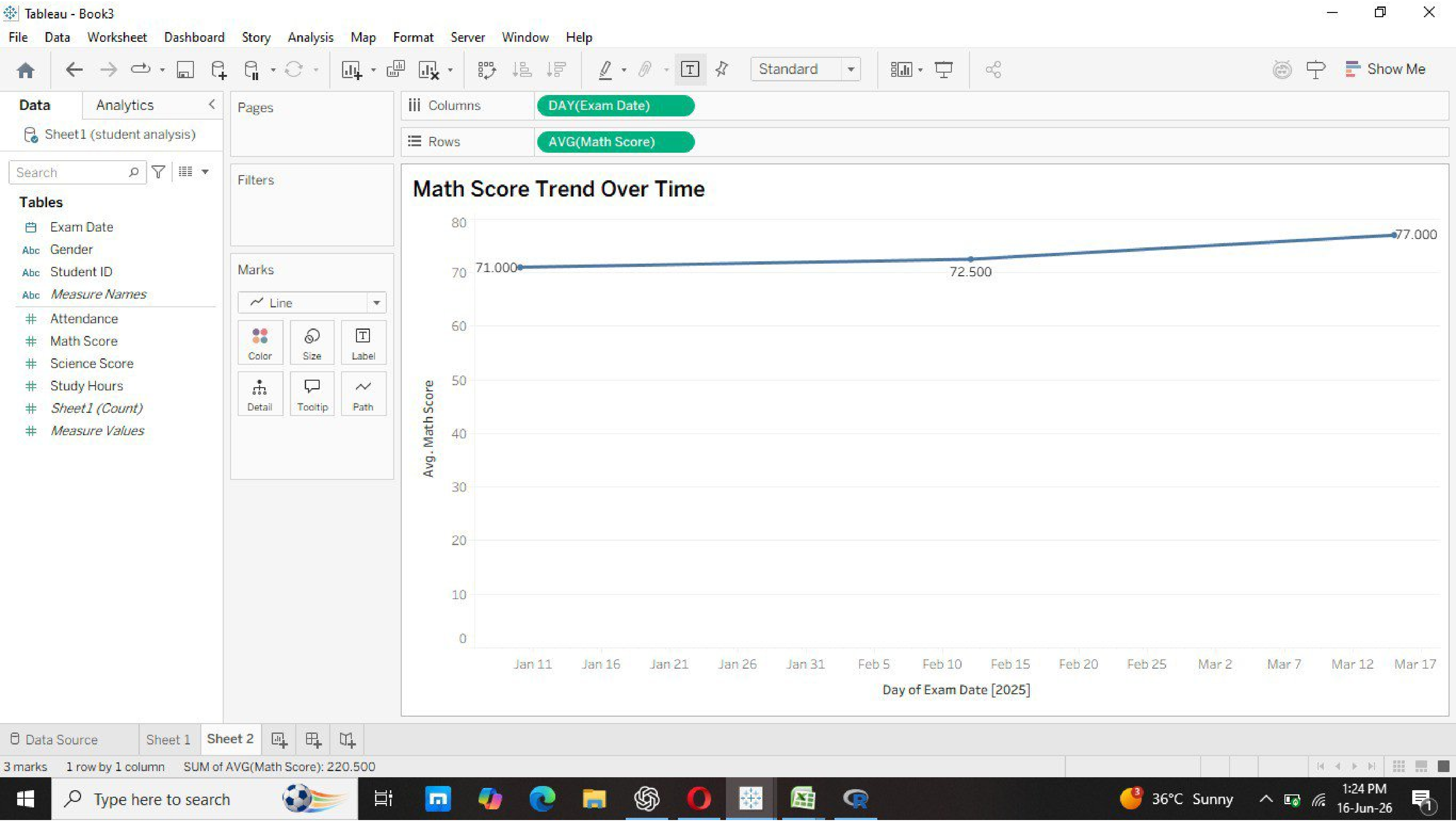


Tableau - Book3
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Show Me
Data Analytics
Sheet1 (student analysis)
Search
Tables
Exam Date
Gender
Student ID
Measure Names
Attendance
Math Score
Science Score
Study Hours
Sheet1 (Count)
Measure Values
Pages
Filters
Marks
Line
Color Size Label
Detail Tooltip Path
Columns DAY(Exam Date)
Rows AVG(Math Score)
Math Score Trend Over Time
80
70
60
50
40
30
20
10
0
Avg. Math Score
71.000
72.500
77.000
Jan 11 Jan 16 Jan 21 Jan 26 Jan 31 Feb 5 Feb 10 Feb 15 Feb 20 Feb 25 Mar 2 Mar 7 Mar 12 Mar 17
Day of Exam Date [2025]
Data Source Sheet 1 Sheet 2
3 marks 1 row by 1 column SUM of AVG(Math Score): 220.500
Type here to search
36°C Sunny
1:24 PM
16-Jun-26

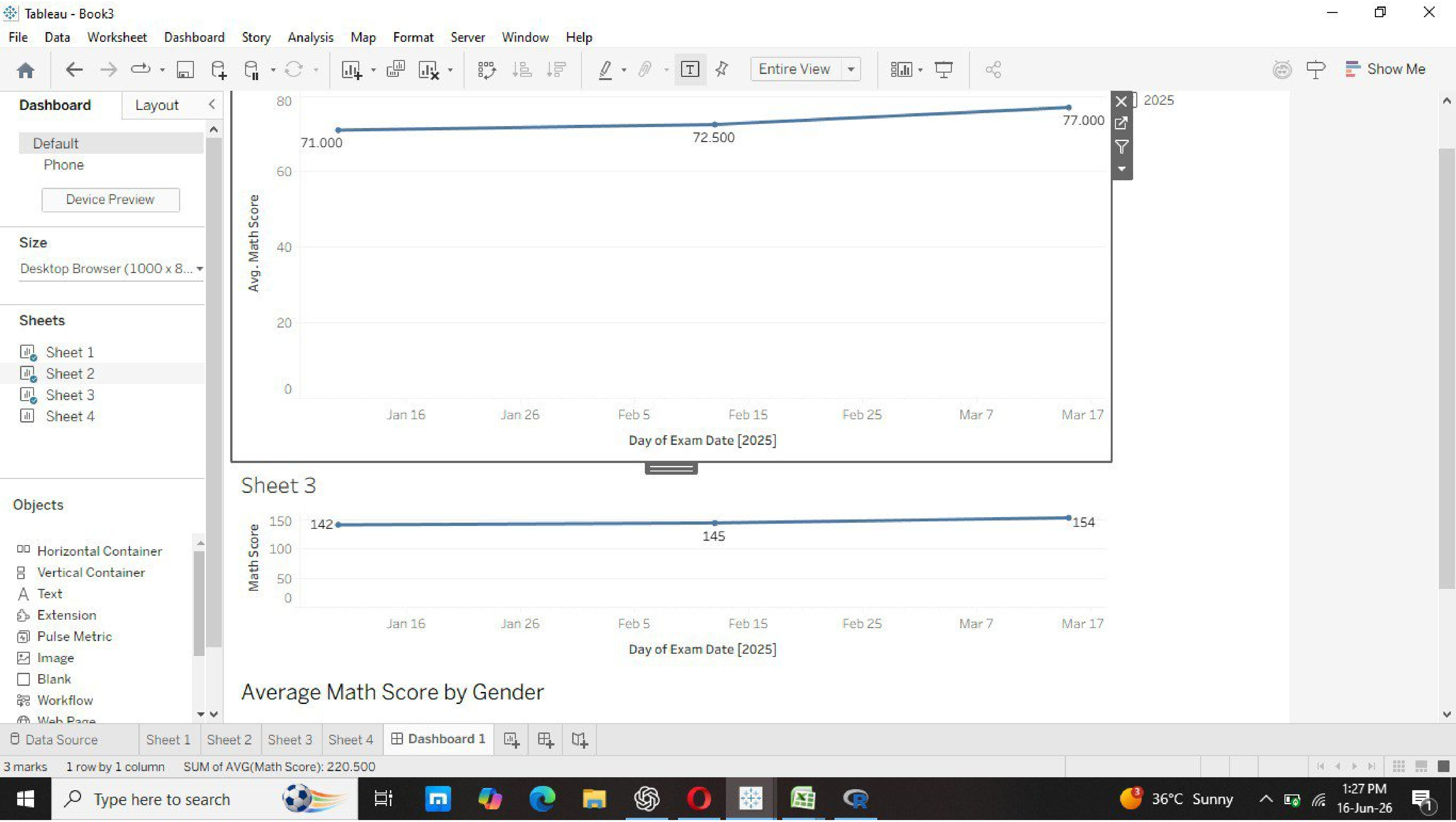


Tableau - Book3
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Entire View
Show Me
Dashboard Layout
Default
Phone
Device Preview
Size
Desktop Browser (1000 x 8...
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Blank
Workflow
Web Page
2025
80
60
40
20
0
Avg. Math Score
71.000
72.500
77.000
Jan 16
Jan 26
Feb 5
Feb 15
Feb 25
Mar 7
Mar 17
Day of Exam Date [2025]
Sheet 3
150
100
50
0
Math Score
142
145
154
Jan 16
Jan 26
Feb 5
Feb 15
Feb 25
Mar 7
Mar 17
Day of Exam Date [2025]
Average Math Score by Gender
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Dashboard 1
3 marks
1 row by 1 column
SUM of AVG(Math Score): 220.500
Type here to search
36°C Sunny
1:27 PM
16-Jun-26